

# Understanding of Generated Standard Parasitic Extracted Format file

## INTRODUCTION:

In this lab, we are going to learn how to extract the SPEF (Standard Parasitic Exchange Format) file from OpenLane. The SPEF file is generated after the routing stage and contains detailed parasitic information such as wire resistance and capacitance. This data is essential for accurate post-layout timing analysis and verification of the digital design. By extracting the SPEF file from OpenLane, we can analyze real interconnect effects and ensure the design meets timing requirements before fabrication.

## APPLICATION:

SPEF extraction in OpenLane is primarily used for post-layout analysis of digital ASIC designs, enabling accurate evaluation of interconnect parasitic effects after placement and routing. It is widely used in timing signoff, power estimation, and signal integrity verification for open-source hardware projects. SPEF files generated from OpenLane support educational labs, research prototypes, RISC-V based designs, and low-cost tapeout flows by providing realistic parasitic data without relying on commercial EDA tools. This process is essential for validating manufacturability and performance in open PDK-based ASIC implementations.

## OBJECTIVE:

After completing this lab, you will be able to:

- Understand the role of SPEF files in post-layout timing and power analysis
- Run the OpenLane flow up to routing to enable parasitic extraction
- Locate and extract the SPEF file from the OpenLane run directory
- Interpret basic contents of a SPEF file such as nets, resistance, and capacitance
- Use the extracted SPEF file for post-layout Static Timing Analysis (STA)

## PROCEDURE TO CREATE A LAB:

### Step 1: Set up OpenLane flow

1.1: Invoke OpenLane in the terminal using the command: openlane

```
coe-2@ip-10-100-19-29:~$ openlane
=====
OpenLane started for user: coe-2
Workspace : /home/coe-2/OpenLaneUser
PDK Root  : /labroot/openlane_pdk
=====
OpenLane Container (ff5509f):/openlane$
```

1.2: Open folder- designs: cd designs

**Note :** Step 2 , 3 & 4 are same as previous lab (Introduction to OpenLane )

**Step 5 : Use the following commands to run the flow:**

# to start complete flow of design

i) flow.tcl -design inverter

```
OpenLane Container (ff5509f):/openlane$ flow.tcl -design inverter
OpenLane v1.0.2 (ff5509f65b17bfa4068d5336495ab1718987ff69)
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Available under the Apache License, version 2.0. See the LICENSE file for more details.

[INFO]: Using configuration in 'designs/inverter/config.json'...
[INFO]: PDK Root: /openlane/pdks
[INFO]: Process Design Kit: skyl30A
[INFO]: Standard Cell Library: skyl30_fd_sc_hd
[INFO]: Optimization Standard Cell Library: skyl30_fd_sc_hd
[WARNING]: DIODE INSERTION STRATEGY is now deprecated; use GRT_REPAIR_ANTENNAS, DIODE_ON_PORTS and RUN_HEURISTIC_DIODE_INSERTION instead.
[INFO]: DIODE INSERTION STRATEGY set to 3. Setting GRT_REPAIR_ANTENNAS to 1
[INFO]: Run Directory: /openlane/designs/inverter/runs/RUN_2026.01.21_10.10.31
[INFO]: Saving runtime environment...
[INFO]: Preparing LEF files for the nom corner...
[INFO]: Preparing LEF files for the min corner...
[INFO]: Preparing LEF files for the max corner...
[INFO]: Running linter (Verilator) (log: designs/inverter/runs/RUN_2026.01.21_10.10.31/logs/synthesis/linter.log)...
[INFO]: 0 errors found by linter
[INFO]: 0 warnings found by linter
[STEP 1]
[INFO]: Running Synthesis (log: designs/inverter/runs/RUN_2026.01.21_10.10.31/logs/synthesis/1-synthesis.log)...
[STEP 2]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/inverter/runs/RUN_2026.01.21_10.10.31/logs/synthesis/2-sta.log)...
[STEP 3]
[INFO]: Running Initial Floorplanning (log: designs/inverter/runs/RUN_2026.01.21_10.10.31/logs/floorplan/3-initial_fp.log)...
[STEP 39]
[INFO]: Running Circuit Validity Checker ERC (log: designs/inverter/runs/RUN_2026.01.21_10.10.31/logs/signoff/39-erc_screen.log)
[INFO]: Saving current set of views in 'designs/inverter/runs/RUN_2026.01.21_10.10.31/results/final'...
[INFO]: Saving runtime environment...
[INFO]: Generating final set of reports...
[INFO]: Created manufacturability report at 'designs/inverter/runs/RUN_2026.01.21_10.10.31/reports/manufacturability.rpt'.
[INFO]: Created metrics report at 'designs/inverter/runs/RUN_2026.01.21_10.10.31/reports/metrics.csv'.
[INFO]: There are no max slew, max fanout or max capacitance violations in the design at the typical corner.
[INFO]: There are no hold violations in the design at the typical corner.
[INFO]: There are no setup violations in the design at the typical corner.
[SUCCESS]: Flow complete.
```

## Step 6: SPEF File Extraction

### 6.1 When is SPEF Generated?

- The SPEF file is generated automatically after routing

### 6.2 Location of SPEF File in Open Lane

After routing completes, navigate to:

```
designs/inverter/runs/RUN_2024.01.21_10.30.45/result/final/spef/inverter.spef
```

```
OpenLane Container (ff5509f):/openlane/designs/inverter/runs/RUN_2026.01.21_10.10.31/results/final/spef$  
/openlane/designs/inverter/runs/RUN_2026.01.21_10.10.31/results/final/spef/inverter.spef  
OpenLane Container (ff5509f):/openlane/designs/inverter/runs/RUN_2026.01.21_10.10.31/results/final/spef$
```

### 6.3 Open vi viewer in openlane

```
Vi. inverter.spef
```

```
1 SPEF "ieee 1481-1999"  
2 *DESIGN "inverter"  
3 *DATE "11:11:11 Fri 11 11, 1111"  
4 *VENDOR "OpenRCX"  
5 *PROGRAM "Parallel Extraction"  
6 *VERSION "1.0"  
7 *DESIGN_FLOW "NAME_SCOPE LOCAL" "PIN_CAP NONE"  
8 *DIVIDER /  
9 *DELIMITER :  
10 *BUS_DELIMITER []  
11 *T_UNIT 1 NS  
12 *C_UNIT 1 PF  
13 *R_UNIT 1 OHM  
14 *L_UNIT 1 HENRY  
15  
16 *NAME_MAP  
17 *3 in  
18 *4 out  
19 *5 net1  
20 *6 net2  
21 *7 FILLER_0_0_21  
22 *8 FILLER_0_0_27  
23 *9 FILLER_0_0_29  
24 *10 FILLER_0_0_41  
25 *11 FILLER_0_0_47  
26 *12 FILLER_0_0_9  
27 *13 FILLER_0_10_15  
:se nu
```

### OBSERVATION:

The OpenLane flow was run successfully up to the routing stage. After routing, the SPEF file was automatically generated and found in the routing directory of the OpenLane run. The presence of the SPEF file confirmed successful parasitic extraction and availability of post-layout interconnect data.

## OUTPUT:

SPEF file: inverter.spf generated in the result directory.

```
*SPEF "ieee 1481-1999"
*DESIGN "inverter"
*DATE "11:11:11 Fri 11 11, 1111"
*VENDOR "OpenRCX"
*PROGRAM "Parallel Extraction"
*VERSION "1.0"
*DESIGN FLOW "NAME_SCOPE LOCAL" "PIN_CAP NONE"
*DIVIDER /
*DELIMITER :
*BUS_DELIMITER []
*T_UNIT 1 NS
*C_UNIT 1 PF
*R_UNIT 1 OHM
*L_UNIT 1 HENRY

*NAME_MAP
*3 in
*4 out
*5 net1
*6 net2
*7 FILLER_0_0_21
*8 FILLER_0_0_27
*9 FILLER_0_0_29
*10 FILLER_0_0_41
*11 FILLER_0_0_47
*12 FILLER_0_0_9
*13 FILLER_0_10_15
"inverter.spf" 178L, 2755C
```

## CONCLUSION:

In this lab, the SPEF file was successfully extracted from the OpenLane flow after the routing stage. The extracted SPEF provided detailed parasitic information including net resistances and capacitances, enabling accurate post-layout timing and power analysis. The analysis confirms that the design's interconnect parasitics are captured correctly, supporting reliable timing verification and signoff. Overall, extracting the SPEF file ensures that the post-layout design data is ready for timing validation and further ASIC verification processes.